

Learning Outcomes

At the end of the session, you will be able to:

- Write, run, and explain logical and relational manipulations.
- Write, run, and explain string operations and perform string manipulations

Activity

1. Write a R program to get weight and height to calculate BMI and display the status as the category below:

BMI	Status
≤ 18.4	Underweight
18.5 - 24.9	Normal
25.0 - 39.9	Overweight
≥ 40.0	Obese

Sample expected output:

```
Underweight: FALSE
Normal: FALSE
Overweight: TRUE
Obese: FALSE
```

```
weight <- readline(prompt="Enter weight in KG: ")
height <- readline(prompt="Enter height in M: ")
weight = as.numeric(weight)
height = as.numeric(height)
BMI = weight/(height**2)
print(paste("Underweight:", BMI<=18.4))
print(paste("Normal:", BMI>=18.5&BMI<=24.9))
print(paste("Overweight:", BMI>=25&BMI<=39.9))
print(paste("Obese:", BMI>=40))
```

2. Write a R program to get two strings from user and compare if the inputs were the same (case-insensitive).

Sample expected output:

```
Enter string 1: test
Enter string 2: exam
This program compare 2 strings. Both inputs are similar: FALSE

Enter string 1: test
Enter string 2: TEST
This program compare 2 strings. Both inputs are similar: TRUE

Enter string 1: Test
Enter string 2: TEST
This program compare 2 strings. Both inputs are similar: TRUE
```

```
str1 <- readline(prompt="Enter string 1: ")
str2 <- readline(prompt="Enter string 2: ")
```

```
str1 = tolower(str1)
str2 = tolower(str2)
cat(paste("This program compare 2 strings. Both inputs are
similar:",str1==str2))
```

3. Write a R program that retrieves from the user the name and phone number. The program will display the name (replace each lowercase letter with the same uppercase letter) and only display the first 3 digits and the last 4 digits from the phone number.

Sample expected output:

```
Hi, CHARLES . A verification code has been sent to 014 -xxxxx 8752
```

```
name <- readline(prompt="Enter name: ")
phone <- readline(prompt="Enter phone no. (012-123456789):")
name<-toupper(name)
front <- substr(phone,1,3)
back <- substr(phone,6,9)
cat(paste("Hi,", name, ". A verification code has been sent to ", front,
"-xxxxx", back))
```

Submission

- Submit to your GA.